NLP HOMEWORK 2

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1 Introduction

Languages change over time, and this makes older texts useless for training LLM for everyday tasks. This project looks at the problem of translating Italian texts from the 13th to 15th centuries into modern Italian. This is more than just swapping words; it requires understanding old grammar and word meanings.

We test two different methods. The first method uses a large language model, which we fine-tune to learn the specific patterns of this translation task. The second method uses a series of two smaller, specialized models. This pipeline translates the old Italian to English, and then from English back to modern Italian.

This report has three main goals. First, we show how to fine-tune a large model like Mixtral-8x7B on standard hardware. Second, we compare this method directly with the two-model pipeline. Finally, we test if other language models can be used as automated judges to score the translations, and we compare their scores to those from a human expert.

2 Methodology

Our experiment was based on two different procedures for translation, using datasets we prepared for training and a detailed system for evaluation.

2.1 Datasets and Preparation

We created two datasets for model training. For fine-tuning the large language model, we prepared a small, high-quality collection called the "Boccaccio" dataset. It includes 18 sentences from Boccaccio's *Decameron* and *Il Filocolo*. A human expert translated these sentences into modern Italian to create a correct reference. The different writing styles in the source texts provided useful training data. For the pipeline method, we created a larger "Dante" dataset by collecting thousands of matching Italian and English verses from the *Divine Comedy*. This dataset was used to train the model that translates archaic Italian to English. For the final evaluation, the test sentences were also translated by a human expert to serve as a correct standard.

2.2 Approach 1: Fine-Tuning a Large Model (Mistral)

This method focused on adapting `Mixtral-8x7B-Instruct-v0.1`, a model chosen because it is good at following instructions. To make it run on our available hardware, we used two important techniques.

First, we used 4-bit quantization to load the model into our 30 GB of GPU memory. With the `BitsAndBytes` library, we used the `nf4` data type, which is designed to store the model's weights more efficiently. We also used double quantization and set the compute type to `bfloat16`. This means that while the model is stored in a memory-saving 4-bit format, all calculations are done in a more precise 16-bit format to maintain accuracy.

Second, we used Low-Rank Adaptation (LoRA) to fine-tune the model without using too much memory. This method keeps the original model numbers frozen and adds very small, trainable layers. We specifically added these LoRA layers to the query (`q_proj`) and value (`v_proj`) parts of the model's self-attention blocks. We set a rank (r) of 8 and a scaling factor (lora_alpha) of 16. This resulted in only about 3.4 million trainable numbers, which made the fine-tuning process manageable.

2.3 Approach 2: Translation Pipeline (Opus)

This procedure used two smaller models from the Helsinki-NLP Opus-MT collection. First, the archaic Italian text was translated to English using `Helsinki-NLP/opus-mt-it-en`, which we fine-tuned on the "Dante" dataset (Figure 1). Second, the English output was translated back to modern Italian using a pre-trained

`Helsinki-NLP/opus-mt-en-it` model.

2.4 Evaluation

We evaluated the results in three ways. First, a human expert scored every translation on a scale of 1 to 5 based on grammar, meaning, and overall quality. Second, we used standard automated scores for translation, such as BERTScore, COMET-22, BLEURT, BLEU, and chrF++. The third way was to use other language models, specifically `Gemini 2.5 Flash` and a 4-bit version of `Unbabel/M-Prometheus-7B`, as automated judges. We then measured how closely their scores matched the human expert's scores using quadratic Cohen's Kappa.

3 Results and Discussion

Our results show a large difference in performance between the two methods. The fine-tuned Mixtral model consistently worked better than the Opus pipeline, as detailed in Table 1.

The human evaluation scores clearly showed that Mixtral performed better. On the general quality score, Mixtral had an average of 3.49, while the Opus pipeline averaged only 2.21. This pattern was the same for the other scores. For grammar, Mixtral scored 4.38 compared to 3.12 for Opus, and for meaning, it scored 4.05 compared to 2.67. This shows the large model was much better at producing translations that were grammatically correct and true to the original meaning.

This was confirmed by the automated scores. As shown in Table 1, Mixtral had better results on all automated tests. For example, its BERTScore was 0.8581, while Opus's was 0.8273. The difference was even larger for scores like COMET-22 (0.7971 vs. 0.6797) and BLEURT (0.6827 vs. 0.5246), which are known to align well with human opinions.

When using other models as judges, the two models behaved very differently. Gemini's scores were very close to the human expert's, giving Mistral an average score of 3.30 and Opus a score of 2.20. In contrast, Prometheus gave higher scores on average, and its scores were much more spread out, often being either very high or very low.

The agreement measurement confirmed this finding. As seen in Table 1, Gemini's scores had a moderate agreement with the human scores, with a quadratic Cohen's Kappa of 0.4467 for Mistral and 0.4454 for Opus. Prometheus, however, showed very little agreement, with a Kappa score of only 0.1614 for Mistral's general quality. This suggests that large, capable models like Gemini can be trusted more to act like human judges for difficult translation tasks.

On a final note, we observed that although the Opus model performed worse overall, its more literal, word-for-word translations sometimes saved parts of the meaning in very difficult sentences. In the same sentences, Mistral sometimes failed by completely changing the sentence and losing the original meaning.

4 Conclusion

This study shows that fine-tuning a large model using special techniques like 4-bit quantization and LoRA is a very good method for translating archaic Italian into its modern form. This method works much better than the pipeline method with smaller models, producing translations with better grammar and more accurate meaning. Our work also shows that modern language models like Gemini can be useful as automated evaluators, which could be a way to evaluate many translations without needing human experts for every one. While the pipeline method was not as good, its more literal translations might be useful if combined with other methods or in situations where it is important to preserve the original words.

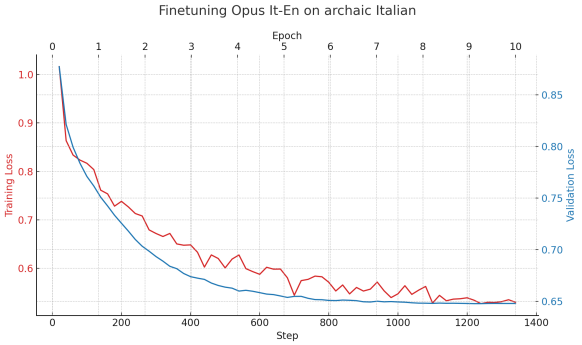


Figure 1: Opus finetuning on "Dante" chart. In Red training loss and in Blue validation loss

A Evaluation Table

Table 1: Evaluation Results for MISTRAL and OPUS Models

Metric	**MISTRAL**			**OPUS**		
	Grammar	**Semantic**	**Generic**	**Grammar**	**Semantic**	**Generic**
Prometheus Mean	3.8866	3.1031	3.9485	3.2165	2.5670	3.4536
Prometheus Std Dev	0.7889	0.5680	0.4648	1.2096	0.7893	1.3230
Prometheus Kappa (q)	0.1151	0.1164	0.1614	0.3611	0.3208	0.0332
Gemini Score Mean	—	—	3.2990	—	—	2.1959
Gemini Score Std Dev	—	—	0.9807	—	—	0.7989
Gemini Kappa (q)	—	—	**0.4467**	—	—	**0.4454**
Human Score Mean	4.3814	4.0515	3.4948	3.1237	2.6701	2.2062
Human Score Std Dev	0.7832	1.1670	1.1468	1.2687	1.4048	0.6281
BERTScore Mean	0.8581	0.8581	0.8581	0.8273	0.8273	0.8273
BERTScore Std Dev	0.0507	0.0507	0.0507	0.0475	0.0475	0.0475
COMET-22 Mean	0.7971	0.7971	0.7971	0.6797	0.6797	0.6797
COMET-22 Std Dev	0.1009	0.1009	0.1009	0.1254	0.1254	0.1254
BLEURT Mean	0.6827	0.6827	0.6827	0.5246	0.5246	0.5246
BLEURT Std Dev	0.1420	0.1420	0.1420	0.1619	0.1619	0.1619
BLEU (corpus)	0.3326	0.3326	0.3326	0.2233	0.2233	0.2233
chrF++ (corpus)	0.5762	0.5762	0.5762	0.4858	0.4858	0.4858
TER (inv, corpus)	0.4737	0.4737	0.4737	0.3625	0.3625	0.3625

B Prompt

B.1 Template

An instruction, a response to evaluate, a **gold translation** (reference answer) and a score rubric representing an evaluation criterion are given.

1. Write a detailed feedback of maximum 100 words that assesses the response strictly according to the score rubric ([SCORE RUBRIC]), **not evaluating in general**. 2. After writing the feedback, write a score that is an integer between 1 and 5. You must refer to the rubric. 3. The output format must be EXACTLY: "<your feedback here> [RESULT] <integer 1-5>" 4. Do not generate any headers, closings, or extra commentary.

###The instruction to evaluate: Traduci il seguente brano di italiano arcaico nel moderno italiano: «ARCHAIC_PASSAGE»

###Response to evaluate (model output): {MODEL_TRANSLATION}

###Gold Translation (reference answer): {GOLD_TRANSLATION}

###Score Rubric (Translation Quality):
5 -
4 -
3 -
2 -
1 -

B.2 Generic rubric

5 – Excellent: The translation is fluid, precise, and natural. It preserves all the semantic nuances of the original text. The syntax is modern yet respectful of the source content. The text is perfectly readable and does not feel machine-generated. Archaic terms are rendered with accurate modern equivalents, even in ambiguous cases. Indistinguishable from a translation by a highly qualified human.
4 – Good: A generally correct and readable translation. The text appears modern and understandable, although some expressions come across as slightly mechanical or overly literal. It preserves the overall meaning well but may lose subtle cultural or historical nuances. Some terms are adapted with a good but not optimal choice. It may resemble a solid human translation, but not an outstanding one.
3 – Sufficient: An understandable translation but with evident mechanical traits or stylistically poor passages. There may be rigid constructions, syntactic calques, or unnatural lexical choices. The global meaning is retained, but the expressive richness of the original medieval text is lost. It reads like an automatic translation with minimal human supervision.
2 – Insufficient: A discontinuous or incoherent translation. Sentences are disconnected, modern vocabulary is misused, and ambiguous or obsolete words are misinterpreted. Some passages are incomprehensible or misleading. It clearly feels like a text translated automatically without human intervention. Not reliable for reading or academic study.
1 – Gravely Insufficient: A failed translation. Sentences are nonsensical, terms are mistranslated or omitted entirely. There is no grammatical or semantic coherence. The text is unusable and devoid of communicative value. Neither the original content nor its purpose can be discerned. It seems produced by a crude or faulty automatic system, utterly lacking linguistic or historical understanding.

B.3 Grammatic Rubric

5 – Flawless: Grammar, punctuation, and orthography are impeccable. Sentences flow naturally, with varied structure and perfect agreement of tense, number, and gender. Word order feels idiomatic and clear.
4 – Very Good: Only rare, minor slips (e.g., a missing comma or slight tense inconsistency). Fluency and correctness remain high; errors do not interrupt reading.
3 – Adequate: Noticeable but non-disruptive issues such as awkward word order, repeated structures, or several punctuation glitches. Readable but needs copy-editing.
2 – Poor: Frequent grammatical errors (agreement mistakes, run-ons, fragments) distract and sometimes blur meaning. Substantial revision required.
1 – Unacceptable: Pervasive, severe faults make the text hard to follow. Basic syntactic rules are ignored; comprehension is unreliable.

B.4 Semantic Rubic

5 – Excellent: All intended meanings, nuances, and relationships are conveyed with precision. Concepts are accurately represented, contextually appropriate, and internally consistent. No misinterpretations or omissions; the text mirrors the source's full informational depth.
4 – Good: The core meaning and almost all subtleties are preserved. A few minor shades of sense or cultural references may be softened, but factual content and logical relations remain intact. Overall

message fidelity is high.
3 – Sufficient: Central ideas are captured, yet several nuances, implications, or culturally loaded elements are diluted or missed. Some statements may feel slightly out of context, but the general message survives without critical distortion.
2 – Insufficient: Major semantic inaccuracies, omissions, or additions alter the author's intent. Key concepts are confused or simplified, leading to a partial or skewed understanding of the original content.
1 – Gravely Insufficient: Meaning is largely lost or contradicted. Essential ideas are misrepresented, context is ignored, and the text may introduce content not present in the source. Readers cannot rely on the translation to grasp the original message.